

## Viterbi-training for HMM (Problem 2(c))

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Observation sequence: HHTTTHHTTTTTHHHH  
Number of epochs: 5

Initial parameters (epoch 0):

$\pi$ :  
0.333333 0.333333 0.333333

A (rows = from state, cols = to state):  
from state 1: 0.800000 0.100000 0.100000  
from state 2: 0.200000 0.700000 0.100000  
from state 3: 0.200000 0.100000 0.700000

B (rows = states 1..3, cols = [H, T]):  
state 1: 0.500000 0.500000  
state 2: 0.700000 0.300000  
state 3: 0.250000 0.750000

Epoch 1

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Best Viterbi path (states): [1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1]  
Joint probability of best path: 1.7895697067e-07

Parameters after epoch 1:

$\pi$ :  
1.000000 0.000000 0.000000

A (rows = from state, cols = to state):  
from state 1: 1.000000 0.000000 0.000000  
from state 2: 0.200000 0.700000 0.100000  
from state 3: 0.200000 0.100000 0.700000

B (rows = states 1..3, cols = [H, T]):  
state 1: 0.500000 0.500000  
state 2: 0.700000 0.300000  
state 3: 0.250000 0.750000

Epoch 2

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Best Viterbi path (states): [1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1]  
Joint probability of best path: 1.5258789062e-05

Parameters after epoch 2:

$\pi$ :  
1.000000 0.000000 0.000000

A (rows = from state, cols = to state):  
from state 1: 1.000000 0.000000 0.000000  
from state 2: 0.200000 0.700000 0.100000  
from state 3: 0.200000 0.100000 0.700000

B (rows = states 1..3, cols = [H, T]):  
state 1: 0.500000 0.500000  
state 2: 0.700000 0.300000  
state 3: 0.250000 0.750000

Epoch 3

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Best Viterbi path (states): [1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1]  
Joint probability of best path: 1.5258789062e-05

Parameters after epoch 3:

pi:  
1.000000 0.000000 0.000000

A (rows = from state, cols = to state):  
from state 1: 1.000000 0.000000 0.000000  
from state 2: 0.200000 0.700000 0.100000  
from state 3: 0.200000 0.100000 0.700000

B (rows = states 1..3, cols = [H, T]):  
state 1: 0.500000 0.500000  
state 2: 0.700000 0.300000  
state 3: 0.250000 0.750000

Epoch 4

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Best Viterbi path (states): [1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1]  
Joint probability of best path: 1.5258789062e-05

Parameters after epoch 4:

pi:  
1.000000 0.000000 0.000000

A (rows = from state, cols = to state):  
from state 1: 1.000000 0.000000 0.000000  
from state 2: 0.200000 0.700000 0.100000  
from state 3: 0.200000 0.100000 0.700000

B (rows = states 1..3, cols = [H, T]):  
state 1: 0.500000 0.500000  
state 2: 0.700000 0.300000  
state 3: 0.250000 0.750000

Epoch 5

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Best Viterbi path (states): [1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1]  
Joint probability of best path: 1.5258789062e-05

Parameters after epoch 5:

pi:  
1.000000 0.000000 0.000000

A (rows = from state, cols = to state):  
from state 1: 1.000000 0.000000 0.000000  
from state 2: 0.200000 0.700000 0.100000  
from state 3: 0.200000 0.100000 0.700000

B (rows = states 1..3, cols = [H, T]):  
state 1: 0.500000 0.500000  
state 2: 0.700000 0.300000  
state 3: 0.250000 0.750000

Final parameters after all epochs:

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Final (same as last epoch):

pi:  
1.000000 0.000000 0.000000

A (rows = from state, cols = to state):  
from state 1: 1.000000 0.000000 0.000000  
from state 2: 0.200000 0.700000 0.100000  
from state 3: 0.200000 0.100000 0.700000

B (rows = states 1..3, cols = [H, T]):  
state 1: 0.500000 0.500000

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state 2: 0.700000 0.300000
state 3: 0.250000 0.750000
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